ECG Simulator Documentation
Part 8: Advanced Reference & Future Roadmap

Extended Features, Phase 4+ Planning, Troubleshooting

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Contents

1 Troubleshooting Guide

1.1 Flat QRS (No Visible QRS Complex)

1.1.1 Symptoms

- ECG shows flat or minimal voltage during QRS interval
- P waves visible (normal)
- T waves visible (normal)
- Only QRS appears depressed or absent

1.1.2 Root Causes

1. **Depol Gaussian decay**: QRS Gaussian has small σ_d (default 12.7 ms)
   - Solution: Increase depol_duration to 40 ms for Phase 3 (LV overlap)
2. **Amplitude scaling too low**: amplitude_scale ¡ 1.0
   - Solution: Set amplitude_scale = 1.0 or higher
3. **Engine selection wrong**: Using Phase 1/2 with Phase 3 parameters
   - Solution: Ensure engine selector matches code logic

1.1.3 Resolution Steps

1. Check GUI amplitude_scale slider (should be 1.0)
2. Verify Engine = "graph" (Phase 1/2) or "cable" (Phase 3)
3. For Phase 3: Ensure depol_duration parameter = 40 ms
4. Click Reset button to clear old traces
5. Click Start to generate new beat

1.2 No ECG Signal Generated

1.2.1 Symptoms

- Click Start button; no ECG trace appears
- Console shows no errors
- Display remains blank

1.2.2 Root Causes

1. **Timer not running**: GUI timer not started
2. **Parameters invalid**: Infinite or negative values
3. **Engine disconnected**: Engine not properly initialized

1.2.3 Resolution

1. Check console output (enable debug_mode = true)
2. Verify HR is between 30–220 bpm
3. Verify all conductances are 0.0–1.0
4. Reset GUI: Click Reset button
5. Try different engine (graph vs cable)

1.3 Irregular Rhythm When Expecting Regular

1.3.1 Causes

1. **AF enabled**: af_enabled = true
   - Solution: Set af_enabled = false
2. **Ectopic focus active**: ectopic_enabled = true
   - Solution: Set ectopic_enabled = false
3. **Escape pacemaker interfering**: escape_enabled = true
   - Solution: Ensure AV conductance = 1.0 so escapes don't fire

2 Advanced Parameter Tuning

2.1 Matching Patient-Specific ECG Patterns

To simulate a patient's ECG with specific morphology:

1. **Measure patient ECG**:
   - HR, PR, QRS, QT intervals
   - QRS axis, QRS amplitude
   - Presence of pathology (bundle block, ischemia, etc.)
2. **Adjust simulator parameters**:
   - Set HR to patient HR
   - Prolonged PR → increase av_delay
   - Broad QRS → reduce lbb_conductance or rbb_conductance
   - Left axis deviation → enable LAHB (set lahb_conductance → 0)
   - Right axis deviation → enable LPHB (set lphb_conductance → 0)
3. **Adjust timing constants** (Phase 3):
   - Modify τ_s in cable_1d.py to match patient's conduction times
   - Re-run numerical search for phase calibration
4. **Validate against patient ECG**
   - Compare intervals and amplitudes
   - Iterate parameter adjustments

3 Phase 4 Development Plan

3.1 Phase 4a: 3-D Ventricular Tissue (Q2 2027)

3.1.1 Objectives

- Extend Phase 3 cable model to 2-D ventricular sheet
- Implement orthotropic diffusion (longitudinal ¿ transverse CV)
- Model 3-D dipole orientation changes during repolarization

3.1.2 Technical Changes

- Replace 1-D Laplacian with 2-D: $\nabla^2 v = \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}$
- Introduce diffusion tensors: $D_L > D_T$ (anisotropy)
- Update CFL condition for 2-D: $\text{CFL} = D_{\max}\Delta t / \min((\Delta x)^2, (\Delta y)^2)$

3.1.3 Expected Outcomes

- Improved QRS morphology (especially V1–V3 precordial leads)
- Realistic T-wave vector changes
- Ability to simulate infarct patterns (regional conduction delay)

3.2 Phase 4b: Full Conduction System (Q3 2027)

3.2.1 Objectives

- Upgrade Phase 3 cable to full 3-D geometry
- Include:
  - SA node (pacemaker dynamics)
  - Complete atrial geometry
  - AV node (two-dimensional with slow-fast regions)
  - His bundle and bundle branches
  - Complete LV and RV

3.2.2 New Parameters

- Fast-pathway conductance (AV node)
- Slow-pathway conductance (dual AV node pathways)
- Accessory pathway (Wolff–Parkinson–White syndrome)

3.3 Phase 5: Advanced Cell Models (Q4 2027)

3.3.1 Courtemanche Model (Atrial)

- 21-variable state vector (vs 2 for FHN)
- Detailed ion channels: I_{Na}, I_K, I_{Ca}, I_{K1}, ...
- APD typically 100–150 ms (clinically accurate)
- Will enable: AF dynamics, APD alternans

3.3.2 ten Tusscher Model (Ventricular)

- 19-variable state vector
- Epicardial, midmyocardial, endocardial variants
- APD typically 200–300 ms
- Will enable: Transmural dispersion, long QT syndrome

3.4 Phase 6: Boundary-Element Forward Model (2028)

3.4.1 Replace Single-Dipole with BE Torso Model

- Construct 3-D torso geometry (from CT/MRI templates)
- Compute BEM gain matrix (heart → surface electrodes)
- Project 3-D ventricular dipoles to 12 standard + additional leads

3.4.2 Expected Improvements

- Realistic lead vectors (St.–axis angles match clinical)
- Ability to model unusual electrode placements
- ST-segment morphology improvements

3.5 Phase 7: Autonomic Nervous System (2028)

3.5.1 Sympathetic/Parasympathetic Modulation

- Model beta-adrenergic effects: increased HR, faster AV conduction
- Model parasympathetic effects: decreased HR, AV block risk
- Parameters: **sympathetic_tone**, **parasympathetic_tone**
- Realistic response to exercise, stress, sleep

3.5.2 Pharmacological Model

- Beta-blockers: Reduce HR, prolong PR
- Calcium channel blockers: AV nodal delay
- Antiarrhythmics: Class I (Na channel), Class II (beta), Class III (K channel)
- Simulate drug effects on conduction and refractoriness

4 Extending the Code

4.1 Adding a Custom Pathology Plugin

4.1.1 Step 1: Create Plugin Class

```
from plugins.base import PluginBase

class MyCustomArrhythmia(PluginBase):
    name = "MyCustomArrhythmia"
    description = "My␣custom␣ECG␣pattern"

    def __init__(self, params):
        self.params = params
        self.enabled = False

    def apply(self, activation_times, conductances):
        if not self.enabled:
            return activation_times, conductances

        # Modify activation times or conductances here
        # Example: block AV node
        conductances['av'] *= 0.5

        return activation_times, conductances
```

4.1.2 Step 2: Register Plugin

```
# In main.py or plugin manager
from plugins.my_custom import MyCustomArrhythmia

engine.register_plugin(MyCustomArrhythmia(params))
```

4.1.3 Step 3: Add GUI Controls

1. Add group box to parameter panel
2. Add enable/disable checkbox
3. Add parameter sliders (e.g., severity, timing)
4. Connect to plugin.apply() method

4.2 Modifying Conduction Geometry

To change the anatomical connectivity (DAG in Phase 1/2):

1. Edit cardiac_sim/core/tissue/graph.py
2. Modify adjacency matrix or edge list
3. Update conduction delays (path_delay dict)
4. Test with BFS algorithm

4.3 Adding New ECG Leads

1. Define new unit vector in cardiac coordinates
2. Add to LEAD_MATRIX in lead_field.py
3. Update lead names list
4. GUI automatically displays new lead

5 Known Limitations

1. **Single-dipole ECG model**: Limited accuracy for far-field effects
2. **No T-wave alternans**: Current model lacks beat-to-beat APD variation
3. **No ischemia simulation**: No regional metabolic effects on conduction
4. **Simplified atrial geometry**: 8-cell cable vs realistic 3-D atria
5. **No autonomic integration**: No HR variability from breathing, stress
6. **FHN cell model**: APD 540 ms (longer than clinical 200–300 ms for ventricle)

6 Quick Reference: Parameter Summary

Parameter	Range	Default	
HR (bpm)	30–220	70	
cycle_length (s)	0.27–2.0	0.857	
av_delay (s)	0.05–0.30	0.15	
av_conductance	0.0–1.0	1.0	
vent_abs_refr (s)	0.25–0.45	0.35	
amplitude_scale	0.1–5.0	1.0	

7 Contact & Support

For bug reports, feature requests, or documentation clarifications:

- **Repository**: [GitHub URL to ECG simulator]
- **Documentation**: See Part 1–7 for comprehensive reference
- **Issues**: File issue on GitHub with reproducible steps
- **Discussion**: Review Phase 4+ planning in this Part 8